

NB-548732

National Board Number

As Required by the Provisions of the ASME Boiler and Pressure Vessel Code Rules, Section VIII, Division 1

1. Manufactured and certified by **NORTHSTAR PROCESS EQUIPMENT, 245 INDUSTRIAL WAY, HOUSTON, TX 77041**
(Name and address of Manufacturer)
2. Manufactured for **BLUE RIDGE ENERGY SYSTEMS, 78 COMMERCE PARK, TULSA, OK 74116**
(Name and address of Purchaser)
3. Location of installation **BAY 4 - TEST FACILITY, FREEPORT, TX**
(Name and address)

4. Type **Vertical** **Separator** **V-26-0417**
(Horizontal, vertical, or sphere) (Tank, separator, jkt. vessel, heat exch., etc.) (Manufacturer's serial number)

K1234.5

(CRN)

DWG-PV-8842 Rev B

(Drawing number)

548732

(National Board number)

2026

(Year built)

5. ASME Code, Section VIII, Div. 1 **2023 / 2025** **CC-2908** **H2S SERVICE**
[Edition and Addenda, if applicable (date)] (Code Case Number) [Special Service per UG-120(d)]

Items 6-11 incl. to be completed for single wall vessels, jackets of jacketed vessels, shell of heat exchangers, or chamber of multichamber vessels.

6. Shell: (a) Number of course(s) **2** (b) Overall length **32' 7.5"**

Course(s)			Material	Thickness		Long. Joint (Cat. A)			Circum. Joint (Cat. A, B, & C)			Heat Treatment	
No.	Diameter	Length	Spec./Grade or Type	Nom.	Corr.	Type	Full, Spot, None	Eff.	Type	Full, Spot, None	Eff.	Temp.	Time
1	5.0" OD	28' 4.0"	SA387-11 CL2	0.625	0.1250	2	Spot	85	2	Spot	85	1275F	2.00HR
2	52.0" x 32.00"	4' 3.5"	SA516-65	0.625	0.1250	2	Spot	85	2	Spot	85	1275F	2.00HR

Body Flanges on Shells										Bolting			
No.	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Location		Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
2	SO	32.00"	38.00"	4.50	0.625	SA350 LF3	Full Pen	Top		40 - 1"-8UNC	SA193-B8M	1.06/2.00	SS316

7. Heads: (a) **SA516-65, 1275F, 2.00HR** (b) **SA516-65, 1275F, 2.00HR**
(Material spec. number, grade or type) (H.T. - time and temp.) (Material spec. number, grade or type) (H.T. - time and temp.)

	Location (Top, Bottom, Ends)	Thickness		Radius		Elliptical Ratio	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure		Category A		
		Min.	Corr.	Crown	Knuckle					Convex	Concave	Type	Full, Spot, None	Eff.
(a)	Top	0.1875	0.1250	48.0	3.0	2:1	30 deg	26.0	0.0	X		1	Spot	85
(b)	Bottom	0.1875	0.1250	48.0	3.0	2:1	30 deg	26.0	0.0	X		1	Spot	85

Body Flanges on Heads										Bolting			
	Location	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached		Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
(a)	Top	SO	32.00"	38.00"	4.50	0.625	SA350 LF3	Welded		40 - 1"-8UNC	SA193-B8M	1.06/2.00	SS316

8. Type of jacket **Dimple** Jacket closure **Welded**
(Describe as type & weld, bar, etc.)

If bar, give dimensions; if bolted, describe or sketch

3/8" x 2" continuous closure bar

9. MAWP **125 psi** **15 psi** at max. temp. **375 F** **350 F** Min. design metal temp. **-10 F** to **125 psi**
(Internal) (External) (Internal) (External)

10. Impact test **Yes, UCS-66** at test temperature of **-10 F**
(Indicate yes or no and the component(s) impact tested)

11. Hydro., pneu., or comb. test pressure **Hydro at 163 PSI** Proof test **150 PSI**

Items 12 and 13 to be completed for tube sections.

12. Tubesheet **SA266 Gr.2** **37.75"** **2.250** **0.1250** **Welded**
(Stationary (material spec. no.)) (Diameter (subject to press.)) (Nominal thickness) (Corr. allow.) Attachment (welded or bolted)

- SA266 Gr.2** **36.25"** **1.750** **0.1250** **Removable**
(Floating (material spec. no.)) (Diameter) (Nominal thickness) (Corr. allow.) Attachment

13. Tubes **SA179** **0.875** **12 BWG** **420** **Straight**
(Material spec. no., grade or type) (O.D.) (Nominal thickness) (Number) [Type (Straight or U)]

Items 14-18 incl. to be completed for inner chambers of jacketed vessels or channels of heat exchangers.

14. Shell: (a) No. of course(s) 2 (b) Overall length 2' 4.0"

Course(s)			Material		Thickness		Long Joint (Cat. A)			Circum. Joint (Cat. A, B, & C)			Heat Treatment	
No.	Diameter	Length	Spec./Grade or Type		Nom.	Corr.	Type	Full, Spot, None	Eff.	Type	Full, Spot, None	Eff.	Temp	Time
2	33.0" OD	2' 4.0"	SA516-65		0.500	0.1250	2	Spot	85	2	Spot	85	1275F	2.00HR

Body Flanges on Shells												
No.	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Location	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
2	SO	32.00"	38.00"	4.750	0.500	SA350 LF3	Full Pen	Top	40 - 1"-8UNC	SA193-B8M	1.06/2.00	SS316

15. Heads: (a) SA516-65, 1275F, 2.00HR (b) SA516-65, 1275F, 2.00HR

(Material spec. number, grade or type) (H.T. - time and temp.)

(Material spec. number, grade or type) (H.T. - time and temp.)

(a)	Location (Top, Bottom, Ends)	Thickness		Radius		Elliptical Ratio	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure		Category A		
		Min.	Corr.	Crown	Knuckle					Convex	Concave	Type	Full, Spot, None	Eff.
(a)	Top	0.2500	0.1250	48.0	3.0	2:1	30 deg	26.0	0.0	X		1	Spot	85

Body Flanges on Heads												
	Location	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
(a)	Top	SO	32.00"	38.00"	4.750	0.500	SA350 LF3	Welded	40 - 1"-8UNC	SA193-B8M	1.06/2.00	SS316

16. MAWP 175 psi 20 psi ax. temp. 450 F 400 F in. design metal temp. -5 F 175 psi
(Internal) (External)17. Impact test Yes, UCS-66 at test temperature of -5 F
(Indicate yes or no and the component(s) impact tested)18. Hydro., pneu., or comb. test pressure Hydro at 228 PSI Proof test 200 PSI

19. Nozzles, inspection, and safety valve openings:

Purpose (Inlet, Outlet, Drain, etc.)	No.	Diameter or Size	Type	Material		Nozzle Thickness		Reinforcement Material	Attachment Details		Location (Insp. Open.)
				Nozzle	Flange	Nom.	Corr.		Nozzle	Flange	
Feed Inlet	2	12	600# RFWN	SA106C	SA182 F316	0.500	0.1250	Pad (d)		Full Pen	12 O'Clock
Product Outlet	2	16	300# RFWN	SA106C	SA182 F316	0.438	0.1250	Pad (d)		Full Pen	6 O'Clock
Vent	2	6	300# RFWN	SA106C	SA182 F316	0.375	0.1250	Integral (d)		Full Pen	Top
Drain	2	3	300# RFWN	SA106C	SA182 F316	0.300	0.1250	Pad (d)		Full Pen	Bottom
PT/LT	4	1	6000# HC	SA182 F316	SA182 F316	0.250	0.1250	None (e)		Threaded	Side

20. Supports: Skirt Yes Lugs 4 Legs Four Others Guide Attached Base Ring Welded
(Yes or no) (Number) (Number) (Describe) (Where and how)

21. Manufacturer's Partial Data Reports properly identified and signed by Commissioned Inspectors have been furnished for the following items of the report (list the name of part, item number, Manufacturer's name, and identifying number):

Channel, tubesheet and nozzle schedule attached as Appendix A.22. Item #19: Nozzle reinforcement per Fig. UG-37.1.PWHT performed per UCS-56.Corrosion allowance applies to pressure-retaining parts.

CERTIFICATE OF SHOP COMPLIANCE

We certify that the statements in this report are correct and that all details of design, material, construction, and workmanship of this vessel conform to the ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. U Certificate of Authorization Number U-41782 Expires 12/31/2027

Date 09/14/2026 Name Northstar Process Equipment Signed A. Reynolds
(Manufacturer) (Representative)

CERTIFICATE OF SHOP INSPECTION

I, the undersigned, holding a valid commission issued by the National Board of Boiler and Pressure Vessel Inspectors and employed by **Hartford Steam Boiler Inspection & Insurance Co.**

have inspected the pressure vessel described in this Manufacturer's Data Report on 09/12/2026, and state that,

to the best of my knowledge and belief, the Manufacturer has constructed this pressure vessel in accordance with ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. By signing this certificate neither the Inspector nor his/her employer makes any warranty, expressed or implied, concerning the pressure vessel described in this Manufacturer's Data Report. Furthermore, neither the Inspector nor his/her employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date 09/14/2026 Signed J. Carter Commissions: A-12345 / N-67890
(Authorized Inspector) [National Board (incl. endorsements)]

CERTIFICATE OF FIELD ASSEMBLY COMPLIANCE

We certify that the statements made in this report are correct and that the field assembly construction of all parts of this vessel conforms with the requirements of ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. U Certificate of Authorization Number _____ Expires _____

Date _____ Name _____ Signed _____
(Assembler) (Representative)

CERTIFICATE OF FIELD ASSEMBLY INSPECTION

I, the undersigned, holding a valid commission issued by The National Board of Boiler and Pressure Vessel Inspectors and employed by _____

have compared the statements in this Manufacturer's Data Report with the described pressure vessel and state that parts referred to as data items _____, not included in the certificate of shop inspection, have been inspected by me and to the best of my knowledge and belief, the Manufacturer has constructed and assembled this pressure vessel in accordance with the ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. The described vessel was inspected and subjected to a hydrostatic test of _____. By signing this certificate neither the Inspector nor his/her employer makes any warranty, expressed or implied, concerning the pressure vessel described in this Manufacturer's Data Report. Furthermore, neither the Inspector nor his/her employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date _____ Signed _____ Commission _____
(Authorized Inspector) [National Board (incl. endorsements)]